Name: Harshita N Kumar
ID: COMETFWC052

avr-gcc Question

Question

For the output F to be 1 in the logic circuit shown, the input combination should be

- a) $A = 1,\ B = 1,\ C = 0$
- b) $A = 1,\ B = 0,\ C = 0$
- c) $A = 0,\ B = 1,\ C = 0$
- d) $A = 0,\ B = 0,\ C = 1$

Question Analysis

From the logic circuit, the Boolean expression for the output is
$F = AB\overline{C}$
The output becomes 1 when
$A = 1$, $B = 1$, and $C = 0$.
Hence the correct option is
$a)\ A = 1,\ B = 1,\ C = 0$

Truth Table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Hardware Implementation

The Boolean expression $F = AB\overline{C}$ is implemented using Arduino UNO and a 7447 decoder connected to a seven segment display.

Required Components

- Arduino UNO
- 7447 Decoder IC
- Common Anode Seven Segment Display
- Breadboard
- Jumper wires
- Resistors

Pin Connections

- Arduino D8 $\rightarrow$ 7447 A
- Arduino D9 $\rightarrow$ 7447 B
- Arduino D10 $\rightarrow$ 7447 C
- Arduino D11 $\rightarrow$ 7447 D
- Seven segment common anode $\rightarrow$ +5V

Logic Description

The Arduino generates the input signals A, B, and C.

The logic expression implemented is

$F = AB\overline{C}$

When the condition $A = 1$, $B = 1$, and $C = 0$ is satisfied, the output becomes 1 and the result is displayed on the seven segment display through the 7447 decoder.

Code Uploading Steps

1. Create a Platform IO project
2. Write the code in main.c in src
3. Run the PIO project with command

```
pio run
```

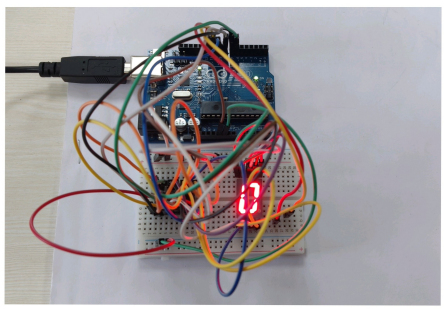
It will compile the code and create the .hex file

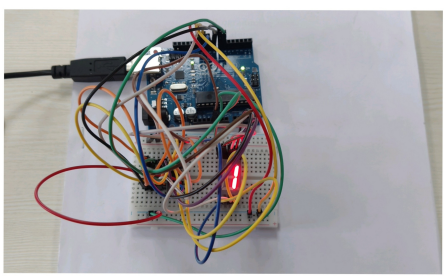
4. Copy the hex file to Arduino Droid folder
5. Connect the Arduino UNO to mobile with OTG cable
6. Upload the hex file using "upload precompiled" option
7. Observe the output and verify the expression

Experimental Truth Table

A	B	C	Output
1	1	0	1

Hardware Setup





Conclusion

The Boolean expression $F = AB\overline{C}$ was verified using Arduino UNO and a seven segment display through the 7447 decoder. The experimental result matches the theoretical analysis.